

Then

$$\bar{\delta} g^{\mu\nu} = g^{\mu\nu}(x) - g^{\mu\nu}(x) = \epsilon(g^{\mu\alpha} K^\nu_{,\alpha} + g^{\alpha\nu} K^\mu_{,\alpha} - g^{\mu\nu} K^\alpha_{,\alpha}) \quad (8.2)$$

$$\bar{\delta} \Lambda = -\epsilon \Lambda_{,\alpha} K^\alpha \quad (8.3)$$

$$\bar{\delta} \phi_{00} = \epsilon R_{i\alpha} K^\alpha_{,i} + \frac{1}{2} \epsilon R_{,,\alpha} K^\alpha \quad (8.4)$$

$$\bar{\delta} \phi_{0i} = \frac{1}{2} \epsilon (R_{i\alpha} K^\alpha_{,3} + R_{3\alpha} K^\alpha_{,i} + R_{i3,\alpha} K^\alpha) \quad (8.5)$$

To obtain the asymptotic group we demand

$$\bar{\delta} g^{ii} = \bar{\delta} g^{12} = \bar{\delta} g^{13} = \bar{\delta} g^{14} = 0$$

$$\bar{\delta} g^{22} = O(\Omega^{-2})$$

$$\bar{\delta} g^{2i} = O(\Omega^{-4})$$

$$\bar{\delta} g^{ij} = O(\Omega^{-6})$$

$$\bar{\delta} \Lambda = O(\Omega^{-7})$$

$$\bar{\delta} \phi_{00} = O(\Omega^{-9})$$

$$\bar{\delta} \phi_{0i} = O(\Omega^{-7})$$

(8.6)

By

$$\bar{\delta} g^{ii} = 0,$$

$$K^i_{,i} = K^{0i}(u, x^i)$$

(8.7)

By $\bar{\delta} \Lambda = O(\Omega^{-7})$

$$K^2 = O(\Omega^{-2})$$

$$\delta g^{12} = 0 = K^2_{,12} + K'_{,1} + g^{23} K'_{,3} + g^{24} K'_{,4} \quad (8.8)$$

$$\therefore K'_{,1} = 0$$

$$K' = K^{01}(x^i)$$

(8.9)

$$\bar{\delta} g^{13} = 0 = K^3_{,12} + g^{33} K'_{,3} + g^{34} K'_{,4}$$

$$\therefore K^3 = K^{03}(u, x^i) - 4\sigma^{-1} P^2 K'_{,3} + O(\Omega^{-3}) \quad (8.10)$$

$$\bar{\delta} g^{23} = O(\Omega^{-4}) = K^3_{,11} + 2\sigma K^3_{,11} + O(\Omega^{-4})$$

$$\therefore K^{03} = K^{03}(x^i)$$

(8.11)

$$\bar{\delta} g^{ij} = g^{i2} K^j_{,2} + g^{j2} K^i_{,2} + g^{i\kappa} K^j_{,\kappa} + g^{j\kappa} K^i_{,\kappa}$$

$$- g^{ij}_{,\kappa} K^\kappa$$

$$\therefore K^{03}_{,4} = -K^{04}_{,3}$$

$$\therefore K^{03}_{,3} = K^{04}_{,4}$$

(8.12)

(8.13)

$$2S K^{03}_{,3} - S_{,3} K^{03} - S_{,4} K^{04} - 2S K' = 0$$

(8.14)

(8.12) and (8.13) imply that K^{0i} is an analytic function of

$\chi^3 + i\chi^4$. This is a consequence of the fact that we have

reduced the leading term of g^{ij} to a conformally flat form. Thus the only allowed transformations of x^i are the conformal transformations of the form:

$$x^3 + ix^4 = \frac{a(x^3 + ix^4) + b}{c(x^3 + ix^4) + d} \quad (8.15)$$

$$ad - bc = 1$$

When the six parameters a, b, c, d are given K' is uniquely determined by (8.14). K^2 is also uniquely determined. Thus the asymptotic group is isomorphic to the conformal group in two dimensions. Sachs (7) has shown that this is isomorphic to the homogeneous Lorentz group. It is also however isomorphic to the group of motions of a 3-space of constant negative curvature which is the group of the unperturbed Robertson-Walker space. Thus the asymptotic group is the same as the group of the undisturbed space. It is not enlarged by the presence of radiation. This is interesting because in the case of gravitational radiation in empty, asymptotically flat space, it turns out that the asymptotic group contains not only the 10 dimensional inhomogeneous Lorentz group, the group of motions of flat space, but also infinite dimensional "supertranslations". It has been suggested that these supertranslations might have some physical significance in elementary particle physics. The above result would seem

to indicate that this is probably not the case since our universe is almost certainly not asymptotically flat though it may be asymptotically Robertson-Walker.

9. What an observer would measure

The velocity vector V_m of an observer moving with the dust will be:

$$V_1 = \Omega^{-1} + O(\Omega^{-5})$$

$$V_2 = \frac{1}{2}\Omega + O(\Omega^{-3})$$

$$V_3 = O(\Omega^{-3})$$

$$V_4 = O(\Omega^{-3})$$

(9.1)

q_m , the projection of the wave vector L^a in the observer's rest-space (the apparent direction of the wave) will be:

$$q_m = \delta_m^2 - V_m V^1$$

$$q_1 = \Omega^{-2} + O(\Omega^{-5})$$

$$q_2 = \frac{1}{2} + O(\Omega^{-4})$$

$$q_3 = 0$$

$$q_4 = 0$$

$$\begin{aligned} q_3 &= O(\Omega^{-4}) \\ q_4 &= O(\Omega^{-4}) \end{aligned}$$

(9.2)

The observer's orthonormal tetrad may be completed by two space-like unit vectors S_m and t_m

$$\begin{aligned} S_1 &= O(\Omega^{-4}) & t_1 &= O(\Omega^{-4}) \\ S_2 &= O(\Omega^{-2}) & t_2 &= O(\Omega^{-2}) \\ S_3 &= \frac{1}{\sqrt{2}} & t_3 &= \frac{-i}{\sqrt{2}} \\ S_4 &= \frac{1}{\sqrt{2}} & t_4 &= \frac{i}{\sqrt{2}} \end{aligned} \quad (9.3)$$

We write $e^\alpha = (V^\alpha, q^a, S^a, t^a)$

By measuring the relative accelerations of neighbouring dust particles, the observer may determine the 'electric' components of the gravitational wave:

$$E_{ab} = -C_{apbq} V^p V^q$$

In the observer's tetrad this has components

$$\begin{aligned}
 E = & \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} O(\Omega^{-4}) + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} O(\Omega^{-4}) \\
 & + \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} O(\Omega^{-5}) + \begin{bmatrix} 0 & 1 & -1 \\ 1 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix} O(\Omega^{-5}) \\
 & + \begin{bmatrix} 1 & 0 & 0 \\ 0 & -\frac{1}{2} & 0 \\ 0 & 0 & -\frac{1}{2} \end{bmatrix} O(\Omega^{-6})
 \end{aligned} \tag{9.4}$$

This should be compared to the behaviour for asymptotically flat space for which

$$\begin{aligned}
 \bar{E} = & \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} O(r^{-1}) + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} O(r^{-1}) \\
 & + \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} O(r^{-2}) + \begin{bmatrix} 0 & 1 & -1 \\ 1 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix} O(r^{-2}) + \begin{bmatrix} 1 & 0 & 0 \\ 0 & -\frac{1}{2} & 0 \\ 0 & 0 & -\frac{1}{2} \end{bmatrix} O(r^{-3})
 \end{aligned} \tag{9.5}$$

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CHAPTER 4

Singularities

If the Einstein equations without cosmological constant are satisfied, a Robertson-Walker model can 'bounce' or avoid a singularity only if the pressure is less than minus one-third the density. This is clearly not a property possessed by normal matter though it might be possessed by a field of negative energy density like the 'C' field. However there is a grave quantum-mechanical difficulty associated with the existence of negative energy density, for there would be nothing to prevent the creation, in a given volume of space-time, of an infinite number of quanta of the negative energy field and a corresponding infinity of particles of positive energy. If we therefore exclude such fields, all Robertson-Walker models must be of the 'big-bang' type, that is they have a singularity in the past and maybe one in the future as well. It has been suggested ¹ that the occurrence of these singularities is a consequence of the high degree of symmetry of the Robertson-Walker models which restricts the expansion and contraction so that they are purely radial and that more realistic models with fewer or no exact symmetries would not have a singularity. This chapter will be devoted

to an examination of this question and it will be shown that provided certain physically reasonable conditions hold, any model must have a singularity, that is, it cannot be a geodesically complete C^1 , piecewise C^2 manifold.

2. The Fundamental Equation

The expansion $\theta = V^a_{;a}$ of a time-like geodesic congruence with unit tangent vector V^a obeys equation (7) of Chapter 2:

$$\theta_{;a} V^a = -\frac{1}{3}\theta^2 - 2\sigma^2 + 2\omega^2 - R_{ab} V^a V^b \quad (1)$$

A point q will be said to be a singular point on a geodesic γ of a time-like geodesic congruence if θ for the congruence is infinite on γ at q . A point q will be said to be conjugate to a point p along a geodesic γ if it is a singular point on γ of the congruence of all time-like geodesics through p . A point q will be said to be conjugate to a space-like hypersurface H^3 if it is a singular point of the congruence of geodesic normals to H^3 . An alternative description of conjugate points may be given as follows: let K^a be a vector connecting points corresponding distances along two neighbouring geodesics in a congruence with unit

tangent vector V^a . Then K^a is 'dragged' along by the congruence, that is

$$\begin{aligned} \mathcal{L}_V K^a &= 0 \\ \therefore \frac{DK^a}{DS} &= V_{a;b} K^b \end{aligned} \quad (2)$$

$$\therefore \frac{D^2 K^a}{DS^2} = R^a_{bcd} V^b V^c K^d \quad (3)$$

Introducing an orthonormal tetrad e^a_m parallelly transported along V^a with $e^a_u = V^a$ we have

$$\frac{d^2 K_m}{ds^2} = -a^u_m K_u \quad (4)$$

where
$$a^u_m = e^u_a e^b_m R_{acbd} V^c V^d$$

A solution of (4) will be called a Jacobi field. There are clearly eight independent solutions. Since V^a and $S^a V^a$ are solutions, the other six independent solutions of (4) may be taken orthogonal to V^a . Then q is conjugate to p along a geodesic γ if, and only if, there is a Jacobi field along γ which vanishes at p and q . This may be shown as follows: the Jacobi fields which vanish at p may be regarded as generating neighbouring geodesics in the irrotational congruence of all time-like geodesics through p . Therefore they obey

$$\frac{dK_m}{ds} = \sum_{m,n} V_{mn} K_n$$

They may be written

$$K_m = A_{mn}(s) \frac{dK_n}{ds} \Big|_p$$

where

$$\frac{dA_{mn}}{ds} = \sum_{m,r} V_{mr} A_{rn}$$

Near p , $A_{mn}(s)$ will be positive definite. There will be a Jacobi field vanishing at p and q if, and only if, $\det(A_{mn}(q)) = 0$

But $A_{mn}(s) = \exp\left(\int_p^s \sum_{m,n} V_{mn} ds'\right)$

Therefore

$$Q = \frac{1}{\det(A_{mn})} \frac{d}{ds} (\det(A_{mn}))$$

and

$$\frac{d^2 A_{mn}}{ds^2} = - \sum_{m,r} a_{mr} A_{rn}$$

Therefore $\frac{d}{ds} (A_{mn})$ is finite

Hence Q is infinite where and only where $\det(A_{mn}) = 0$.

Thus the two definitions of conjugate points are equivalent.

This also shows that singular points of congruences are points where neighbouring geodesics intersect.

For null geodesic congruences with parallelly transported tangent vector ℓ^a we may define the convergence ρ as in Chapter 3. This obeys

$$\rho_a \ell^a = \rho^2 + \sigma \bar{\sigma} + \frac{1}{2} R_{ab} \ell^a \ell^b \quad (5)$$

We define a singular point of a null geodesic congruence as one where ρ is infinite.

The condition that the pressure is greater than minus one-third the density may be stated more generally as condition (a).

(a) $E > 0$, $E > \frac{1}{2} T$, for any observer with 4-velocity ω^a , where $E = T_{ab} \omega^a \omega^b$ is the energy density in the rest-frame of the observer and $T = T^a_a$ is the rest-mass density.

Condition (a) will be satisfied by a perfect fluid with density $\mu > 0$ and pressure $p > -\frac{1}{3}\mu$. It implies $R_{ab} V^a V^b > 0$ for any time-like or null vector V^a . Therefore by equations (1) and (5) any time-like or null irrotational geodesic congruence must have a singular point on each geodesic within a finite affine distance. Obviously if the flow-lines form an irrotational geodesic congruence, there will be a physical singularity at the singular points of the congruence where the density and hence the curvature are infinite. This will be the case if the universe is filled with non-rotating dust ^{2,3}. However, if the flow-lines are not geodesic (ie. non-vanishing pressure gradient) or are rotating, equation (1) cannot be applied directly.

3. Spatially Homogeneous Anisotropic Universes

The Robertson-Walker models are spatially homogeneous and isotropic, that is, they have a six parameter group of motions transitive on a space-like surface. If we reduce the symmetry by considering models that are spatially homogeneous but anisotropic (that is, they have a three parameter group of motions transitive on a space-like hypersurface) then the matter flow may have rotation, acceleration and shear. Thus there would seem to be the possibility of non-singular models. L. Shepley⁴ has investigated one particular homogeneous model containing rotating dust and has shown that there is always a singularity. Here a general result will be proved.

There must be a singularity in every model which satisfies condition (a) and,

(b) there exists a G_r of motions on the space or on universal covering space^{*}, $r \geq 3$ which is transitive on at

* See section 5

least one space-like surface but space-time is not stationary,

(c) the energy-momentum tensor is that of a perfect fluid,

$T_{ab} = (\mu + p) u_a u_b - p g_{ab}$. u^a is the tangent to flow-lines and is uniquely defined as the time-like eigenvector of the Ricci tensor.

PROOF

R , the curvature scalar must be constant on a space-like surface of transitivity H^3 of the group. Therefore $R_{;a}$ must be in the direction of the unit time-like normal V^a to H^3 .

$$V_a = \frac{e(R_{;a}) R_{;a}}{\sqrt{f}}$$

where $R_{;a} R^{;a} = f > 0$

$e(R_{;a})$ is an indicator = +1 if $R_{;a}$ is past directed
= -1 if $R_{;a}$ is future directed

Then $V_{[a;b]} = 0$.

Thus V_a is a congruence of geodesic irrotational time-like vectors. By condition (a), $R_{ab} V^a V^b > 0$.

Therefore the congruence must have a singular point on each geodesic (by equation 1) either in the future or in the past.

Further, by the homogeneity, the distance along each geodesic from H^3 to the singular point must be the same for each geodesic.

Thus if the surfaces of transitivity remain space-like, they must degenerate into, at the most, a 2-surface C^2 which will be uniquely defined. Let M be the subset of the flow-lines of the matter which intersect C^2 . Let L be the non-empty subset of H^3 intersected by M . Since there is a group transitive on H , L must be H^3 itself. Thus all the

flow-lines through H^3 must intersect the 2-surface C^2 . Thus the density will be infinite there and there will be a physical singularity. Alternatively if the surfaces of transitivity do not remain space-like, there must be at least one surface which is null - call this S^3 . At S^3 , $\rho = 0$, $R_{;a} \neq 0$ (if $R_{;a}$ is zero, we can take any other scalar polynomial in the curvature tensor and its covariant derivatives. They cannot all be zero if space-time is not stationary). We introduce a geodesic irrotational null congruence on S^3 with tangent vector L^a where $L_a R_{;a}$. Then by equation (5), there will be a singular point of each null geodesic in S^3 within finite affine distance either in the future or in the past. The 2-surface of these singular points will be uniquely defined. The same argument used before shows that the density becomes infinite and there is a physical singularity. In fact as S^3 is a surface of homogeneity, the whole of S^3 will be singular and it is not meaningful to call it null or to distinguish this case from the case where the surfaces of transitivity remain space-like.

The conditions (a), (b), (c) may be weakened in two ways. Condition (b) that there is a group of motions throughout space-time may be replaced by (b') and (d).

(b) There is a space-like hypersurface H^3 in which there are three independent vector fields χ^a_m such that

$$\chi^a_m g_{bc} = 0, \quad \chi^a_m R_{bcde} = 0 \quad \text{on } H^3.$$

That is, there is one homogeneous space section.

(d) There exist equations of state such that the Cauchy development of H^3 is determinate.

Then succeeding space-like surfaces of constant R are homogeneous and much the same proof can be given that there are no non-singular models satisfying (a), (b), (c), (d).

The only property of perfect fluids that has been used in the above proof is that they have well defined flow-lines intersection of which implies a physical singularity. Obviously, however, this property will be possessed by a much more general class of fluids. For these, we define the flow vector as the time-like eigenvector (assumed unique) of the energy-momentum tensor. Then we can replace condition (c) on the nature of the matter by the much weaker condition (e).

(e) If the model is singularity-free, the flow-lines form a smooth time-like congruence with no singular points with a line through each point of space-time.

Condition (e) will be satisfied automatically if conditions (a) and (c) are.

This proof rests strongly on the assumption of homogeneity which is clearly not satisfied by the physical universe locally though it may hold on a large enough scale. However it would seem to indicate that large scale effects like rotation cannot prevent the singularity.

It is of interest to examine the nature of the singularity in the homogeneous anisotropic models since this is more likely to be representative of the general case than that of the isotropic models. It seems that in general the collapse will be in one direction,⁵ that is, the universe will collapse down to a 2-surface. Near the singularity, the volume will be proportional to the time from the singularity irrespective of the precise nature of the matter. It also appears that the nature of the particle horizon is different. There will be a particle horizon in every direction except that in which the collapse is taking place.

4. Singularities in Inhomogeneous Models

Lifshitz and Khalatnikov⁶ claim to have proved that a general solution of the field equations will not have a singularity. Their method is to contract a solution with a singularity which they claim is representative of the general solution with a singularity, and then show that it has one fewer arbitrary function than a fully general solution.

Clearly their whole proof rests on whether their solution is fully representative and of that they give no proof. Indeed it would seem that it is not representative since it involves collapse in two directions to a 1-surface whereas in general one would expect collapse in one direction to a 2-surface. In fact their claim has been proved false by Penrose⁷ for the case of a collapsing star using the notion of a 'closed trapped surface'. A similar method will be used to prove the occurrence of singularities in 'open' universe models.

5. 'Open' and 'Closed' Models

The method used by Penrose to prove the occurrence of a physical singularity depends on the existence of a non-compact Cauchy surface. A Cauchy surface will be taken to mean a complete, connected space-like surface that intersects every time-like and null line once and once only. Not all spaces possess a Cauchy surface: examples of those that do not include the plane-wave metrics,⁸ the Godel model,⁹ and N.U.T. space.¹⁰ However none of these have any physical significance. Indeed it would seem reasonable to demand of any physically realistic model that it possess a Cauchy surface. If the Cauchy surface is compact, the model is commonly said to be 'closed'; if non-compact, it is said to

to be 'open'. The surfaces, $t = \text{constant}$, in the Robertson-Walker solutions for normal matter are examples of Cauchy surfaces. If $K = -1$, they have negative curvature and it is frequently stated that they are non-compact. This is not necessarily so: there exist possible topologies for which they are compact. However, the following statements may be made about the topology of the surfaces $t = \text{constant}$.

If the curvature is negative, $K = -1$, the universal covering space is non-compact and is diffeomorphic to E^3 .¹¹ Any other topology can be obtained by identification of points. Thus any other topology will not be simply connected and, if compact, must have elements of infinite order in the fundamental group. Further if compact, they can have no group of motions.¹²

If the curvature is zero, $K = 0$, the universal covering space is E^3 . There are eighteen possible topologies.¹³ If compact they have a G_3 of motions and Betti numbers, $B_1 = 3$, $B_2 = 3$.¹²

If the curvature is positive, $K = +1$, the universal covering space is S^3 . Thus all topologies are compact. The Betti numbers are all zero.¹²

Since a singularity in the universal covering space implies a singularity in the space covered, Penrose's method is applicable not only to spaces that have a non-compact Cauchy

surface but also to spaces whose universal covering space has a non-compact Cauchy surface. Thus it is applicable to models which, at the present time, are homogeneous and isotropic on a large scale with surfaces of approximate homogeneity which have negative or zero curvature.

6. The Closed Trapped Surface

Let T^3 be a 3-ball of coordinate radius r in a 3-surface H^3 ($t = \text{const.}$) in a Robertson-Walker metric with $K = 0$ or -1 . Let q^a be the outward directed unit normal to T^2 , the boundary of T^3 , in H^3 and let V^a be the past directed unit normal to H^3 . Consider the outgoing family of null geodesics which intersect T^2 orthogonally. At T^2 , ρ , their convergence will be:

$$\rho = \frac{1}{2} (V_{a;b} + q_{a;b}) (S^a S^b + t^a t^b)$$

where S^a t^a are unit space-like vectors in H^3 orthogonal to q^a and to each other,

therefore
$$\rho = \frac{2}{R} \left[\sqrt{\frac{\mu}{3} - K} - \frac{1}{r} \sqrt{1 - Kr^2} \right]$$

If $\mu > 0$ and $K = 0$ or -1 , by taking r large enough, we may make ρ negative at T^2 . Therefore, in the language of Penrose, T^2 is a closed trapped surface.

Another way of seeing this is to consider the diagram

in which the flow-lines are drawn at their proper spatial distance from an observer. They all meet in the singularity at $t = 0$. If the past light cone of the observer is drawn on this diagram, it initially diverges from his world-line ($\rho < 0$). It reaches a maximum proper radius ($\rho = 0$) and then converges again to the singularity ($\rho > 0$). The intersection of the converging light cone and the surface H^3 gives a closed trapped surface T^2 . If the red-shift of the quasi-stellar 3C9 is cosmological then it will be beyond the point $\rho = 0$ if we are living in a Robertson-Walker type universe with normal matter. However, the assumptions of homogeneity and isotropy in the large seem to hold out to the distance of 3C9. Thus there is good reason to believe that our universe does in fact contain a closed trapped surface. It should be pointed out that the possession of a closed trapped surface is a large scale property that does not depend on the exact local metric. Thus a model that had local irregularities, rotation and shear but was similar on a large scale at the present time to a Robertson-Walker model would have a closed trapped surface.

Following Penrose it will be shown that space-time has a singularity if there is a closed trapped surface and :

- (f) $E \gg 0$ for any observer with velocity
- (g) there is a global time orientation

(h) the universal covering space has a non-compact Cauchy surface H^3 .

PROOF

Assume space-time is singularity free. Let F^4 be the set of points to the past of H^3 that can be joined by a smooth future directed time-like line to T^2 or its interior T^3 . Let B^3 be the boundary of F^4 . Local considerations show that $B^3 - T^3$ is null where it is non-singular and is generated by the outgoing family of past directed null geodesic segments which have future end-point on T^2 and past end-point where or before a singular point of the null geodesic congruence. Since at T^2 , the convergence, $\rho > 0$ and since $R_{ab}L^aL^b > 0$ by (f), the convergence must become infinite within finite affine distance. Thus $B^3 - T^3$ will be compact being generated by a compact family of compact segments. Hence B^3 will be compact. Penrose's method is then as follows: approximate B^3 arbitrarily closely by a smooth space-like surface and project B^3 onto H^3 by the normals to this surface. This gives a many-one continuous mapping of B^3 into H^3 . Since B^3 is compact, its image B^{3*} must be compact. Let $d(Q)$ be the number of points of B^3 mapped to a point Q of H^3 . $d(Q)$ will change only at the intersection of caustics of the normals with H^3 . Moreover, by continuity $d(Q)$ can only change by an even number.

But $\alpha(\tau^3) = 1$ since this is the identity and $\alpha(H^3 - B^{3*}) = 0$.

This is a contradiction, thus the assumption that space-time is non-singular must be false. An alternative procedure which avoids the slightly questionable step of approximating B^3 by a space-like surface is possible if we adopt condition (e) on the nature of the matter, then B^3 may be projected continuously one-to-one onto H^3 by the flow-lines. This again leads to a contradiction since B^3 is compact and H^3 is not.

7. Cauchy Horizons

~~In the above proof it was necessary to demand that H^3 be a Cauchy surface otherwise the whole of B^3 might not have been projected onto H^3 . We will define a semi-Cauchy surface (s.C.s) as a complete connected space-like surface that intersects every time-like and null line at most once. A s.C.s H^3 will be a Cauchy surface for points near it, that is, it will intersect every time-like and null line through these points. However, further away there may be regions for which it is not a Cauchy surface. Let P^4 be the set of points for which H^3 is a Cauchy surface and let Q^3 be the boundary of these points. Q^3 , if it exists, will be called the Cauchy Horizon relative to H^3 . Q^3 must be a null surface. Furthermore if condition (f) holds the null geodesics~~

generating $\mathcal{Q}^3$ must have at least one singular point which means that they must be bounded in at least one direction. A simple example of s.c.s with a Cauchy horizon is a space-like surface of constant negative curvature completely within the null cone of the point in Minkowski space. The null cone forms the Cauchy horizon.

If conditions (e) and (g) hold, then a model with a compact s.c.s, H^3 must have the topology: $H^3 \times E^1$.

PROOF

Suppose there were a region V^4 through which there were no flow lines intersecting H^3 , then V^3 the boundary of V^4 must be a time-like surface generated by flow-lines which do intersect H^3 . Proceeding along these flow-lines in the direction of their intersection with H^3 , we must reach an end-point of the generator since V^3 does not intersect H^3 . But the existence of this end-point contradicts (e) since it implies a singularity of the flow-line congruence. Thus V^4 must be empty and every point has a flow line through it intersecting H^3 . Thus we have a homeomorphism of the space to $H^3 \times E^1$ by assigning to every point of the space its distance along the flow-line from H^3 and the point of intersection of the flow line with H^3 . It can also be shown that

~~in this case H^3 must be a Cauchy surface. For, suppose there were a Cauchy horizon Q^3 ; this can intersect each flow-line at most once. Therefore there is a homeomorphism of Q^3 into H^3 . Further, by (e) every flow-line must intersect Q^3 . Thus Q^3 is homeomorphic to H^3 and is compact. If condition (f) holds every null geodesic generator of Q^3 has at least one end point. This must be in the direction away from H^3 since in the direction towards H^3 each generator must be unbounded. This however is impossible since Q^3 is compact. Thus H^3 is a Cauchy surface.~~

8. Singularities in 'Closed' Universes

There is a singularity in every model which satisfies (a), (g) and (i).

(i) There exists a compact Cauchy surface H^3 whose unit normal V^a has positive expansion everywhere on H^3 .

PROOF

For the proof it is necessary to establish a couple of lemmas. Assume that space-time is singularity-free. The following result is quoted without proof, it may readily be derived from lemmas proved in reference 11.

If p and q are conjugate points along a geodesic γ and x is a point on γ not in pq then x must have a conjugate point in pq .

An immediate corollary is that if q is the first point along γ conjugate to p and y is in pq then y has no conjugate points in pq . Also since the result that x has a conjugate point in pq can only depend on the values of Q in pq , any irrotational geodesic congruence including the geodesic γ must have a singular point on γ in pq . Thus if q is a point on M^3 and γ is the geodesic normal to M^3 through q , then a point conjugate to q along γ cannot occur until after a point conjugate to M^3 .

If M^3 is a complete connected space-like surface which intersects every time-like and null line from a point p , we may define a function over M^3 as the square of the geodesic distance from p which is taken as positive if the geodesic is time-like and negative if the geodesic is space-like. We call this the world function σ with respect to p . For the closed set of values $\sigma \geq 0$, σ will be a continuous (in general multi-valued) function over M^3 . A time-like geodesic

γ from p will be said to be critical if it corresponds to a value of σ for which $\sigma_{;\mu} \epsilon^\mu_i = 0$ ($i = 1, 2, 3$) where ϵ^μ_i are three independent vectors in M^3 . Clearly a critical geodesic must be orthogonal to M^3 . A geodesic

which is critical will be said to be maximal if it corresponds to a local maximum of .

Lemma 1.

A geodesic γ cannot be maximal for a smooth M^3 if there is a point X conjugate to M^3 but no point conjugate to q on γ in qp , where q is the intersection of γ and M^3 .

Let f_m and g_m be the Jacobi fields along γ which vanish at X and p respectively. They may be written

$$f_m = A(s) f_{mn}^n,$$

$$g_m = B(s) g_{mn}^n.$$

Then $h^n \left(\frac{dA_{mn}}{ds} \Big|_q - \frac{dB_{mn}}{ds} \Big|_q \right) h_n$ must be positive for any h since if it were negative for any h by taking $a = -b h h_{mb} h_{mn}$ beyond q , it would be possible to have a point y on γ beyond q conjugate to X before a point conjugate to p . If it were zero X would be conjugate to p . This shows that the surface at q of constant geodesic distance from p lies nearer to p in every direction than the surface of q of constant geodesic distance from X does. Since X is conjugate to M^3 the surface at q of constant geodesic distance from p lies closer to p in some direction than M^3 does. Hence γ is not maximal.

must be positive definite at x . But at x

$$\begin{pmatrix} A & B \\ m & n \end{pmatrix} \begin{pmatrix} \delta' & \delta' \\ m & n \end{pmatrix} = \begin{pmatrix} B & \delta' & \delta' \\ m & m & n \end{pmatrix} \begin{pmatrix} \delta' & \delta' \\ m & n \end{pmatrix} < 0$$

Thus $\frac{d}{ds} \begin{pmatrix} A & B \\ m & n \end{pmatrix}$ cannot be positive definite at q . Therefore there must be some direction $\begin{pmatrix} m \\ n \end{pmatrix}$ for which

$$\begin{pmatrix} m & n \\ K & K \end{pmatrix} \frac{d}{ds} \begin{pmatrix} A \\ m & n \end{pmatrix} < \begin{pmatrix} m & n \\ K & K \end{pmatrix} \frac{d}{ds} \begin{pmatrix} B \\ m & n \end{pmatrix}$$

and

$$\begin{pmatrix} m & n \\ K & K \end{pmatrix} V < \begin{pmatrix} m & n \\ K & K \end{pmatrix} W$$

at q , where $\begin{pmatrix} m \\ n \end{pmatrix}$ is the unit tangent vector of the congruence of geodesics through p . Thus in the direction K the surface of constant geodesic distance from p lies closer to p than the surface M^3 does. Therefore γ is not maximal.

If M^3 is compact or if the intersection of all time-like and null lines with M^3 is compact, σ must have a maximum value, thus there must be a geodesic normal to M^3 through p longer than γ . We use this to prove another lemma.

Lemma 2

If p lies to the future (past) on a time-like geodesic γ through q , beyond a point z conjugate to q , and there exists a compact Cauchy surface H^3 through q , then there must be another time-like geodesic from p to q longer than γ .

Let y be the last point conjugate to q on γ before p . Let x be the nearest point to p conjugate to p in pq . Let r be a point in yx . Let K^3 be the set of points which have a future (past) directed geodesic of length rq from q . Then

K^3 will be a space-like hypersurface through r . Let F^4 be the set of points which have at least one future (past) directed geodesic from q of length greater than rq . Then the boundary of F^4 , $J^3 \subset K^3$. Since p is in F^4 and since every past (future) directed time-like and null line from p intersects H^3 which is not in F^4 , they must also intersect J^3 . Let L^3 be the intersection of J^3 and these lines. Since H^3 is compact, L^3 must be compact. Consider the function σ with respect to p over K^3 . Its maximum must lie in the compact region L^3 . But, by the previous lemma γ is not maximal, moreover, local considerations show that a singular point in the surface J^3 cannot be a maximum of σ . Thus the maximum value of σ must occur for a geodesic from p orthogonal to L^3 . This must also be a geodesic from p to q of length greater than γ .

Using these two lemmas the theorem may be proved. Since the future (past) directed normals to H^3 are converging everywhere on H^3 , there must be a point conjugate to H^3 a finite distance along each future (past) directed geodesic normal. Let β be the maximum of these distances. Let p be a point on a future (past) directed geodesic normal at a distance greater than β . Consider the function σ with

respect to p over the compact surface H^3 . Let λ be the geodesic from p normal to H^3 at the point q , where σ has its maximum. There must be a point conjugate to H^3 along λ in qp . But if there is no point conjugate to q along λ in qp , then λ cannot be maximal by the first lemma. If however there is a point conjugate to q along λ in qp , then there must be a longer geodesic from q to p by the second lemma. Thus λ is not the geodesic of maximum length from H^3 to p . This is a contradiction which shows that the original assumption that the space was non-singular must be false.

This proof could also be used to show the occurrence of a singularity in a model with a non-compact Cauchy surface provided that the expansion of its normals was bounded away from zero and provided that the intersection of the Cauchy surface with all the time-like and null lines from a point was compact.

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